

ASHWIN KUMAR RANGARAJAN

+1-(864)-449-9085 | ashwintwister@gmail.com | Irvine, CA 92612

[LinkedIn](#) | [Portfolio](#)

Controls and systems integration engineer with 6+ years of experience developing, validating, and integrating vehicle motion control systems, including steer-by-wire, brake-by-wire, torque vectoring, semi-active suspension, MIL/SIL/HIL, ECU bring-up, CAN/LIN/UDS diagnostics, and safety-critical in-vehicle validation.

Education

- Master of Science in Automotive Engineering** – Vehicle Dynamics & Controls
Clemson University - International Centre for Automotive Research **Aug '23 – Aug '25**
Greenville, SC, USA
- Bachelor of Engineering in Mechanical Engineering**
SSN College of Engineering **Aug '14 – May '18**
Chennai, Tamil Nadu, India

Professional Experience

Systems Test & Integration Engineer

Rivian Volkswagen Group Technologies – Vehicle Motion/Propulsion Controls

Nov '25 – Present

Irvine, CA, USA

- Owned feature-level and regression validation for vehicle motion and propulsion control software across R1 and EDV platforms, supporting customer OTA releases, merge releases, and supplier ECU firmware updates.
- Translated vehicle dynamics control requirements into SIL, HIL/Dyno HIL, and in-vehicle verification activities, aligning test execution with program milestones and release-readiness criteria.
- Validated CAN/LIN/UDS and Automotive Ethernet interfaces while automating Python-based test plans and post-processing workflows, maintaining traceability through JAMA, JIRA, GitLab CI/CD, and quality metrics.
- Collaborated with controls, software integration, and supplier teams to validate ECU interface changes, triage vehicle-level issues, and support software release sign-off.

Chassis Controls Software & Systems Intern

Ford Motor Company - Research & Advanced Engineering Division

May '25 – July '25

Dearborn, MI, USA

- Led deployment and real-time validation of a dual-redundant Steer-by-Wire platform from concept through integration, executing MIL, HIL, and proving-ground verification for motion-control functionality and robustness.
- Developed and integrated real-time Steer-by-Wire control software on dSPACE MicroAutoBox ECU, validating CAN interfaces, redundancy behavior, calibration changes, and vehicle response.
- Supported prototype bring-up and hardware/software debugging using oscilloscopes, digital multimeters, and power supplies across bench, HIL, and in-vehicle environments.

Deep Orange 16 - Vehicle Dynamics & Controls Engineer / Graduate Research Assistant

Clemson University – ICAR

Jan '24 – Apr '25

Greenville, SC, USA

- Developed physics-based vehicle dynamics control strategies including Direct Yaw Moment Control, supervisory brake-by-wire logic, and active suspension validation for a high-voltage 4-wheel hub-motor vehicle.
- Designed and validated MIL/HIL setups for brake-by-wire and active suspension systems, mapping ISO 26262-oriented fault responses to test cases, fault-injection scenarios, and fail-safe behavior.
- Led vehicle instrumentation, sensor integration, and dynamic testing on a BMW 535xi using oscilloscopes, DMMs, power supplies, and CAN/IMU/accelerometer/ride-height data to validate and refine system models.

Assistant Manager, Product Development – Chassis Testing & Validation

Royal Enfield (A unit of Eicher Motors Pvt. Ltd.),

Aug '18 – Jun '23

Chennai, Tamil Nadu, India

- Managed chassis systems integration and verification activities, resolving development issues through benchmarking, cross-functional collaboration, root-cause analysis, and validation of design changes.
- Led ABS calibration and component-/vehicle-level test planning using dynamic vehicle data, supporting repeatable verification workflows and compliance alignment with ECE, FMVSS, and Indian standards.

Skills

- MATLAB/Simulink/Stateflow, Embedded Coder, CarSim, Python, C/C++, Embedded C, PowerShell, Bash, WSL/Linux; Automotive CAN/CAN FD, LIN, Ethernet protocols, Unified Diagnostic Services (UDS), XCP; Vector CANalyzer, CANape, CANdb++ , ASAP2 Studio, dSPACE, ATI Vision/VDA; MIL/SIL/HIL, ECU bring-up, Test automation, OTA validation, GitLab/GitHub, CI/CD pipelines, Jenkins, SCons/Bazel fundamentals, JAMA, JIRA, ISO 26262 fundamentals.